

SHENG ZHAO

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My research interest focus on designing efficient and trustworthy interaction methods between **human** and **XR** to enhance human well-being.

EDUCATION

Tsinghua University, Beijing, China 2021.09 – Present
B.Sc. in **Fundamental Science (Physics)** & B.Eng. in **Electrical Engineering** Dual degree
• GPA: 3.81/4; Major GPA: 3.86/4.
• A+: Optics, Quantum Mechanics; A: Electromagnetism, Thermodynamics, Calculus, Mathematical Physics Equations, etc.

PUBLICATIONS & MANUSCRIPTS

- **Sheng Zhao**, JunRay Zhu, Xueyang Wang, Hongyi Li, Fang Yi, Hewu Li, Shuning Zhang and Xin Yi[†]. CoordAuth: A Two-factor Authentication Method in Virtual Reality Leveraging Head-Eye Coordination. *Ubicomp'24*, under review. [PDF](#).
- Xueyang Wang*, **Sheng Zhao***, Yihe Wang, Ziyu Han, Xin Tong and Xin Yi[†]. EmoBlend: Revealing the Impact of Facial Blendshape Intensity on Emotional Perception in Virtual Reality. *TVCG*, under review.
– * indicates equal contribution. [†] denotes the corresponding author.

RESEARCH EXPERIENCES

EmoBlend: Revealing the Impact of Facial Blendshape Intensity on Emotional Perception in VR 2023.04 – 2024.01
Research Assistant, Advisor: *Xin Yi, Yuanchun Shi* Tsinghua University
My contributions:
• Implementing VR face2face chat using TCP-based socket and Oculus-Movement sdk in Unity.
• Designing both qualitative and quantitative research methods based on interviews and questionnaires, conducting user studies.
• Proposing an improved camera2blendshape method to mitigate the uncanny valley effect and enhance the emotional experience.
Submitted to *TVCG*. [Details](#).

CoordAuth: A Two-factor Authentication Method in VR Leveraging Head-Eye Coordination 2023.10 – 2024.01
Research Assistant, Advisor: *Xin Yi* Tsinghua University
My contributions:
• Project leader, proposing the current two-factor authentication system based on unlock pattern and behavior biometrics.
• Using Unity to build the unlock UI leveraging gaze interaction in VR.
• Proposing a voting mechanism using two types of classifiers implemented by random forest to achieve behavioral biometric authentication factor and implementing the algorithm by python.
• Conducting user studies and the paper work.
Rejected by *IMWUT'24-a*. In submission to *CHI'25*. [PDF](#). [Details](#).

AvatarMVFusion: Text-Guided Multi-View Diffusion for Avatar Generation 2024.11 – Present
Research Assistant, Advisor: *Bowen Zhou* Tsinghua University
My contributions:
• Fine-tuning diffusion model by dreambooth for avatar generation with high quality. [Details](#).
• Testing the performance of Zero123 (SOTA) and DreamFusion, figuring out the importance of prompt-driven multi-view generation for the avatar leveraging diffusion.
• Proposing a novel text-guided denoising 3D Unet framework and conducting experiments to evaluate its performance.
Preparing for NIPS'24.

SKILLS

- Program languages: C/C++/C#, Python, Java, MATLAB and Verilog.
- Tools and Frameworks: Git, LaTeX, Unity, PyTorch, Scikit-learn, Markdown and MSP430.
- Techniques: VR development, Quantitative and Qualitative research.

COMPETITION AWARDS AND HONORS

- *First Prize*, 36th China Physics Olympiad (CPHO), 0.9% 2020
- *First Prize*, The 12th Tsinghua University Digital System Design Competition, 1.7% 2022
Designed a small car driven by MSP430 with various sensors. [Details](#).
- *Second Prize*, Tsinghua University Software Design Competition 2023
Developed a Smart Home app using Java. [Code](#).
- *First Prize*, The 3rd Tsinghua Craftsman Competition, 1%, presented a speech 2023
- *First Prize*, Tsinghua University Technological Innovation Scholarship 2023
- *First Prize*, Tsinghua University Sports Excellence Scholarship 2022, 2023